



CASE STUDY

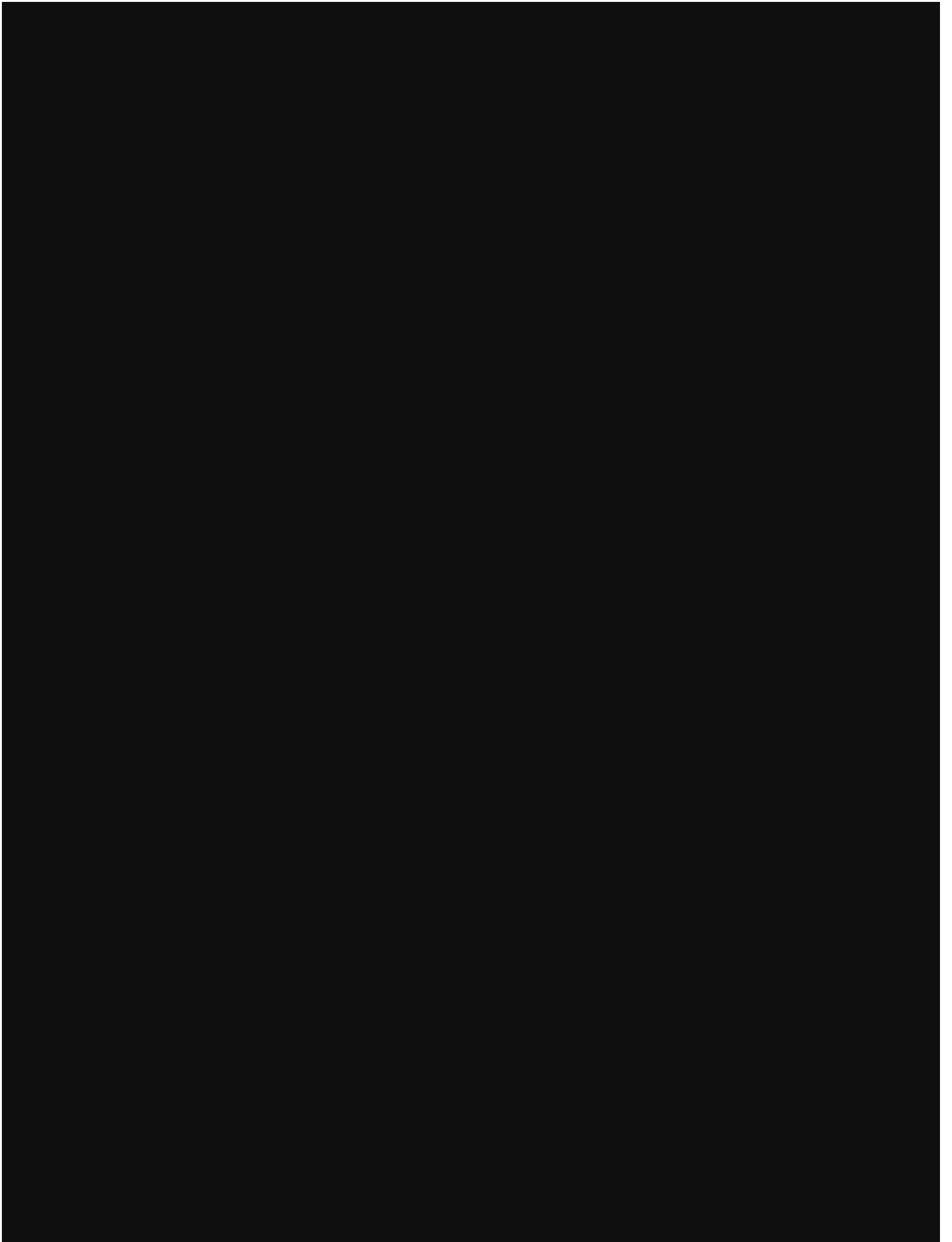
PrivacyHub: Reimagining YouTube's Data Consent under India's DPDP Act, 2023

Empowering users with transparent, purpose-based consent while preserving YouTube's personalized experience.



Piyush Bhardwaj

Vedantu Product Management Internship Assignment



Problem Statement & Context

DPDP ACT CHALLENGES

Core Problem

YouTube relies heavily on user data to drive recommendation accuracy. Under India's **DPDP Act, 2023**, legacy models of generic, bundled consent are no longer compliant. We must transition to **explicit, granular, and revocable** consent without degrading user engagement or causing conversion drop-offs.

Opportunities

Transforming a legal compliance mandate into user trust. By demonstrating high transparency and granular user control, YouTube can improve its user sentiment and set a new standard for Indian consumer tech.

Data Collected & Purposes

- **Watch & Search History:** Powers personalized recommendation feeds.
- **Device Info & Location:** Used for local language & regional content curation.
- **Advertising Interactions:** Delivers higher advertiser relevance and ROI.
- **Voice Searches:** Stores voice logs to improve accessibility search accuracy.

DPDP Notice Mandate: Consent request must specify what data is being collected and exactly what purpose it will serve.

User Persona: Rahul

TARGET AUDIENCE PROFILE



Rahul, 19
College Student • Daily
YouTube User

Goals: Find academic coding guides quickly; stream music & entertainment.

Motivations: High-relevancy feed recommendations save search energy.

Frustrations: Intrusive, irrelevant advertisements; complex settings menus.

Privacy Concerns: Dislikes feeling "watched"; wants transparent explanation of location & voice history storage.

Rahul's Empathy Map

SAYS

"Just let me view my video tutorials. Why do you need my live coordinates?"

THINKS

"Does YouTube record my ambient microphone sounds? How do I delete this data?"

DOES

Skips privacy banners without reading; accesses settings only when highly annoyed.

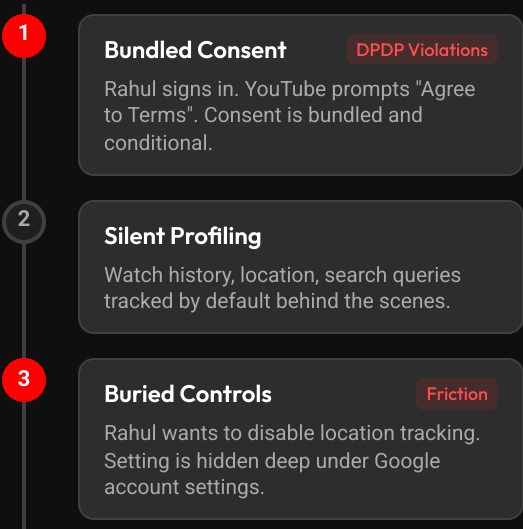
FEELS

Dependent on recommendations, but slightly anxious about Google's data profiling.

Current Consent Experience

USER PAIN POINTS & COMPLIANCE RISKS

Rahul's Current Consent Journey



⚠ DPDP Non-Compliance Audit

- **No Granular Control:** User cannot consent to Watch History personalization while declining Location tracking. Consent is binary ("Take it or leave it").
- **Incomprehensible Terms:** Privacy policies are long, complex text files, violating DPDP requirement for clear notice in plain language.
- **Buried Consent Revocation:** DPDP specifies withdrawing consent must be as easy as giving it. Currently, revocation requires a 5-step external navigation.

Proposed Solution: PrivacyHub

MATERIALIZING DPDP
REQUIREMENTS

The Four Product Pillars

1. Purpose-Based Consent

Granular opt-ins for specific use cases (watch history, location, ads) with plain-language explanations.

2. PrivacyHub Dashboard

Self-service dashboard within Settings. Allows users to view status, download data, or delete accounts.

3. Child Protection

Verifiable parental consent for under-18 users. Block behavioral ads and location tracking for child accounts.

4. Just-in-Time Prompts

Contextual privacy cards explaining location and microphone needs at point of action (e.g. voice search clicks).

UX Comparison

Legacy Experience	PrivacyHub Experience
One generic consent (bundled)	Explicit, purpose-based consent
Settings buried deep in Google Account	Dedicated central PrivacyHub in YouTube
Revocation requires multiple steps	One-tap consent withdrawal
No simple way to purge account logs	Self-serve download and deletion requests
Opaque, legal-only warnings	Layered Just-in-Time explanations

Why This Solution?

MULTI-STAKEHOLDER BENEFITS

Users Gain

- **Transparency:** Layered visual descriptions instead of complex legal terms.
- **Granular Control:** Choice to accept watch history but block location coordinates.
- **Simple Revocation:** Central self-service data management panel.

Business Gains

- **Legal Compliance:** 100% adherence to India's DPDP Act, 2023.
- **Customer Trust:** Transparency acts as a brand differentiation booster.
- **Relevance Preservation:** Encourages watch history opt-in for feed relevance.

Engineering Gains

- **Modular Consent:** Microservice-driven layout decouples consent logic.
- **Reusable Consent APIs:** Can be leveraged across other Google properties in India.
- **Auditable Event Logs:** Streamlined logs simplify future regulatory audit calls.

Product Metrics

TRACKING PRIVACY AND ENGAGEMENT

NORTH STAR METRIC

Measures compliance footprint across Indian active base

97.8%

Valid Active Consent Rate (% of MAU)

ACQUISITION

Signup Conv. Rate >98%

98.4%

Onboard Drop-off

1.6%

COMPLIANCE

Consent Comp.

99.1%

Verifiable Parental

91.4%

ENGAGEMENT

Recommend CTR

12.2%

Daily Active (DAU)

180M

PRIVACY CONTROL

Dashboard Visits

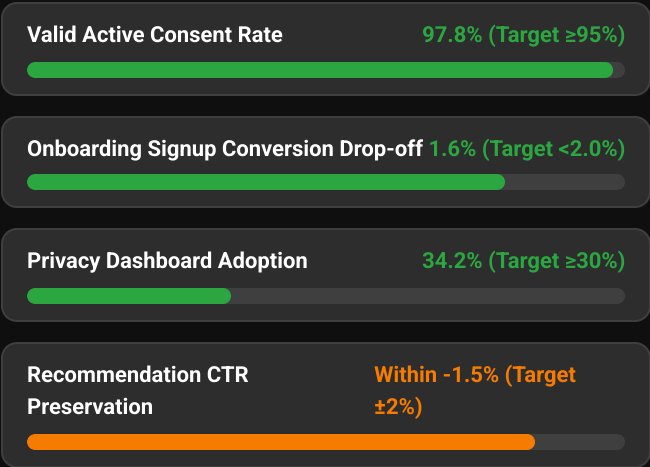
34.2%

Withdraw Rate

4.8%

Success Criteria & Guardrails

MEASURABLE TARGET METRICS



Legal Compliance
Zero Audit Failures

Support Efficiency
-25% Privacy Tickets

CONSENT RATE HYPOTHESIS TREE



A technical crash/loading delay occurs in the client-side consent prompt.

Deploy network retries for slow 3G connection regions.

15% drop in valid active consent base (MAU).

Warning copy regarding recommendation degradation causes users to dismiss.

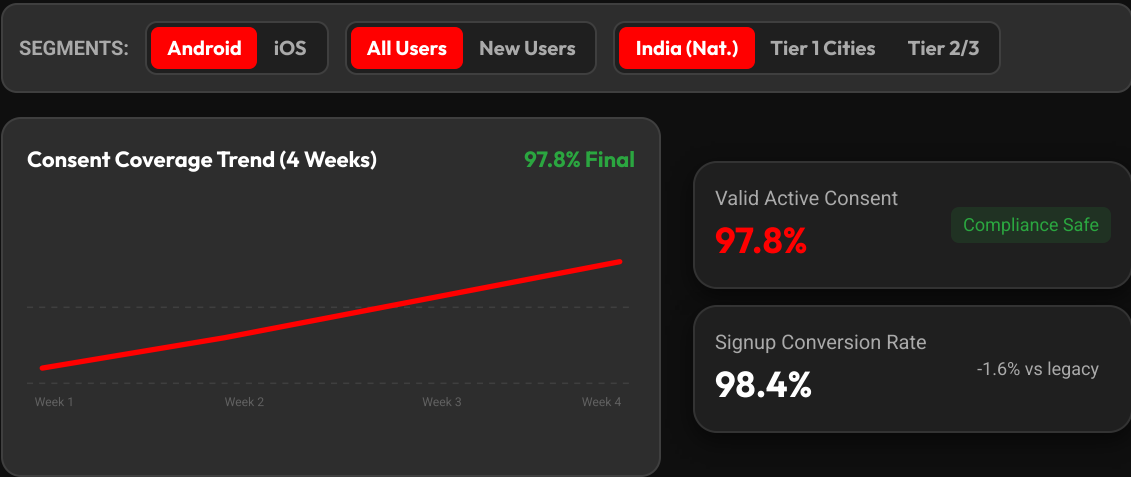
A/B test UX copy adjustments highlighting benefit of history tracking.

Consent changes are recorded in the client UI but fail to commit to databases.

Fix consent backend logger latency dropouts.

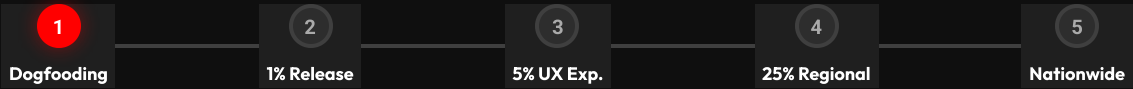
PM Analytics Dashboard

LIVE SEGMENTED PERFORMANCE



Rollout Strategy & Trade-offs

IMPLEMENTATION TIMELINE & RISK CONTROL



Product Decision	Primary Benefit	Trade-off & Cost
Granular Consent Switches	Full transparency, DPDP alignment	Slightly increased signup friction (+1.6% drop-off)
Integrated PrivacyHub Dashboard	Empowered user control & self-serve deletion	Significant backend event logger engineering effort
Just-in-Time Voice/Location Prompts	Contextual relevancy, lower immediate friction	Increased local state caching complexity in app
Under-18 Ad Blocking & Parental Verify	Protects minors, full legal compliance	Forgo targeted ad-monetization for child accounts

Key Takeaways

EXECUTIVE SUMMARY

Summary Roadmap

Problem: Bundled consent violates DPDP 2023, while optional profiling opt-outs risk degrading feed relevance and ad revenue.

Solution: PrivacyHub framework introduces Purpose-Based Consent, contextual Just-in-Time prompts, and a centralized control dashboard.

Impact: Achieves full regulatory compliance, reduces support tickets by 25%, and maintains feed relevance with $\pm 2\%$ recommendation CTR stability.

97.8% North Star: Valid Active Consent Rate

<2.0% Guardrail: Onboarding signup conversion drop

$\geq 30\%$ Adoption: PrivacyHub dashboard visits

